



Wasim Ansari

AI/ML Engineer | Gen AI Solution Architect | IITM Alumnus

✉ wsmaisys@gmail.com

📞 7253923415

🚩 Indian

📍 Dehradun, India

🌐 LinkedIn Profile

🐙 GitHub

🏆 Kaggle

🔗 Portfolio

📄 Publications

Summary

AI/ML Engineer with legal, retail and global business expertise, delivering production-ready Generative AI and Machine Learning solutions. Experienced in legal RAG assistants, price prediction systems, and LLM-powered interview simulators using Mistral, LangChain, Streamlit, Docker, and GCP. IIT Madras-trained, with 7+ years of cross-domain insight to design scalable, business-aligned AI systems.

Projects

AI Travel Planner (Multi-Agent Orchestration using CrewAI with online search tool) [🔗](#)

- **Summary:** Built a multi-agent AI orchestration where three coordinated agents autonomously research tourist destinations, analyze user preferences, and generate personalized end-to-end travel plans.
- **Technologies:** CrewAI, LangChain, Streamlit, Python, CI/CD.

MediFlow - Medical AI Assistant (Multi-Agent Chatbot with MCP RAG tool and custom tools) [🔗](#)

- **Summary:** A medical assistant chatbot that delivers personalized post-discharge care through multi-agent workflow. Receptionist with patient history tool & Clinical Agent with Nephrology RAG and search tools.
- **Technologies:** LangChain, LangGraph, LangSmith, MCP client, Google Cloud Run, CI/CD, HTML, React js.

Nephrology RAG MCP Server - exposing Nephrology RAG services as streamable HTTP [🔗](#)

- **Summary:** Designed an MCP server exposing domain-specific retrieval-augmented generation (RAG) services as streamable HTTP endpoints for seamless integration with multiple AI agents concurrently.
- **Technologies:** FAISS, MCP Server, Python, Asyncio, Google Cloud Run, Docker, JSON-RPC 2.0

DocuDroid - AI Document Analysis Bot (LLM + Vector Search for Smart Document QA) [🔗](#)

- **Summary:** Implemented a temporary in-memory vector store for immediate retrieval, combined with LLM-generated answers. Useful to chat instantly with the PDFs or static documentation webpages.
- **Technologies:** Python, FastAPI, LangChain, InMemoryVectorStore, Docker, Google Cloud Run.

PDF Excel Extractor - AI-Power-Tool (extracts data from PDFs and provided structured Excel) [🔗](#)

- **Summary:** LLM analyzes documents to determine relevant fields for extraction (no hardcoded schema). Converts extracted text into structured Excel spreadsheets for downstream task. Complete Automation.
- **Technologies:** PDF Scraping, LLM schema detection, FastAPI, BLEU Score, Docker, Google Cloud Run.

Flux - Text-to-Image Generator (Powered by Hugging Face Diffusion Model) [🔗](#)

- **Summary:** A powerful image generation tool leveraging state-of-the-art HuggingFace diffusion models to create images from text prompt. Offers intuitive UI as well as API to connect with other services.
- **Technologies:** Python, azurewebsites, Hugging Face Diffusers, PIL, Docker, FastAPI.

Jurisol - Legal AI Assistant (LLM based RAG application with online search and summarization tools) [🔗](#)

- **Summary:** Engineered an intelligent legal search assistant using Retrieval-Augmented Generation (RAG), Chromadb vector database. Equipped with summarization tool and online legal document search.
- **Technologies:** RAG, Vector Database, LangGraph, FastAPI, Mistral LLM, Document Summarization.

Mogambo - Voice AI Calendar Assistant (Voice enabled AI Agent) [🔗](#)

- **Summary:** Developed an AI calendar assistant with voice and text interaction, integrated with Google OAuth. Implemented context-aware scheduling, intent clarification, and automated workflows.
- **Technologies:** Google Calendar Automation, LangGraph, Google OAuth, TTS, STT, FastAPI.

BikeValuePro - ML Model for price prediction (Machine Learning Model with 94.9% Accuracy)

- **Summary:** Trained a ML model for used bike prices using Scikit-learn and XGBoost. Includes smart Streamlit UI and deployment-ready architecture. Display personalized advice based on predicted price.
- **Technologies:** Machine Learning, Predictive Analytics, XGBoost, Scikit-learn, Pandas, Streamlit.

IITM Assignment Solver (Problem solving using Code Generation and Self Executing AI Agent)

- **Summary:** Built an AI agent to process ZIP, CSV, PDF, Excel, and JSON files, generate Python code, and run it in a sandboxed environment. Developed to participate in an IITM college challenge securing A+ Grade.
- **Technologies:** AI Agent, Code Generation, LLM, File Processing, FastAPI, LangChain, Mistral, Automation.

Top Skills

Data Analysis & Exploration

- Statistical Analysis (Python, Pandas, NumPy, Metabase, OpenRefine, Hypothesis Testing)
- Visualization (Matplotlib, Seaborn, PowerBI)

Generative AI & Agentic AI

- Frameworks (LangChain, LangGraph, CrewAI, Agno, MCP clients and servers)
- Vector Databases (ChromaDB, FAISS, Qdrant)
- Workflow Orchestration & Automation (n8n)

Cloud & DevOps Platforms

- Google Cloud (Cloud Run, Cloud SQL, Cloud Storage, Cloud Build, SDK)
- Azure (Container Apps, App Service / Web Apps)
- Vercel, GitHub Pages, Hugging Face Spaces

Machine Learning & Deep Learning

- ML Algorithms (Supervised & Unsupervised)
- Neural Networks (ANN, CNN, RNN, Transfer Learning, Transformers, GAN, Diffusion)
- Frameworks (Scikit-Learn, TensorFlow, Keras, PyTorch, PyTorch Lightning)

MLOps & Model Deployment

- Model Serving (Docker, FastAPI, Flask, Streamlit)
- CI/CD & Versioning (Git, GitHub Actions)

Monitoring & Observability

- Model Performance Tracking (Evidently AI, Grafana, TrackIO, LangSmith)
- Drift Detection & Automated Retraining
- Guardrailing, Prompt Experimentation

Experience

09/2024 – Present
Dehradun, India

WSMAISYS (Independent Freelancer)

- AI/ML Engineer & Gen AI Solution Architect
- Pursuing **IIT Madras' Diploma in Data Science** while independently advancing skills in Transformers, Reinforcement Learning, Generative AI, and Cloud-based ML systems.
 - Developed AI solutions, including **GenAI applications**, predictive **modeling pipelines**, **computer vision** models, voice AI agents, **multi-agent coordination systems**, **RAG-based chatbots**, and custom **AI power tools** tailored for real-world use cases.
 - Delivered production-ready applications using **Docker**, **FastAPI**, **Flask**, **Streamlit**, and **Google Cloud/Azure** with a focus on scalability, robustness, and secure deployment practices.
 - Ranked in the **Top 5% on Kaggle** in a cybersecurity detection challenge, demonstrating strong applied machine learning capabilities and practical problem-solving.
 - Actively building a **diverse project portfolio** to demonstrate adaptability, technical depth, and eagerness to take on real-world AI/ML challenges.

12/2022 – 09/2023
Kingston Upon Hull,
UK

Booze Blitz - Ahaan Limited

- Retail Sales Manager
- Led compliance and strategic sales planning that boosted revenue by 20%.
 - Inspired ML-based solutions for pricing, demand forecasting and **retail analytics**.

09/2021 – 11/2022
Dehradun, India

Ahlan Exports

- Import Export Entrepreneur
- Managed international B2B logistics, compliance and digital marketing funnels.
 - Experience translates into AI interest in **supply chain optimization**.

02/2015 – 01/2020 Dubai, UAE	FAHZAM Properties Real Estate Consultant - Managed property portfolios & client legal clearances. Certified RERA Broker. - Insights now inform AI projects in real estate valuation and docs automation.
09/2013 – 04/2014 Hyderabad, India	QuisLex Legal Services Legal Associate - Reviewed legal documentation & case summaries with accuracy and compliance. - Inspired the need for automation in document review and compliance QA via AI.
12/2012 – 07/2013 Pune, India	Mindcrest India Research Associate-Citator - Classified legal cases with 90%+ accuracy in half the average processing time. - Domain insights now power AI-based legal summarizers and DB management.

Education

09/2024	Indian Institute of Technology, Madras ↗ Diploma in Data Science Pursuing Coursework: ML, Linear Algebra, Statistics, Business Analytics, Tools for Data Science, Deep Learning, Gen AI, Capstone Projects. Grade: CGPA 8+ Projects: System Threat Forecaster (Top 5% in Kaggle), Assignment AI Agent
2007 – 2012	Savitribai Phule Pune University ↗ Bachelor of Laws (5 years integrated) Coursework: Constitution, Civil and Criminal Laws, Procedural Laws, IPR, Banking Law, Environment Law, Drafting. Grade: Higher Secondary Class Achievements: Participant of 1st Student Parliament, Moot Court Competitions.

Languages

English — Fluent IELTS 7 Band	Hindi — Native/Bilingual Native or Bilingual
Urdu — Native/Bilingual Native or Bilingual	

Certifications

Understanding Google Cloud Platform ↗	Azure Essentials ↗	Dubai RERA Certification ↗
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Thoughts and Publications

Adapting Export Strategies for Indian Handmade Carpets: A Response to Rising Global Tariffs ↗ Discover a data-driven export strategy for India's handmade carpet industry navigating rising U.S. tariffs with advanced Python-led trade analytics, market opportunity scoring, and competitive benchmarking.
Hallucinations in AI Legal Research: A Lawyer's Worst Nightmare? ↗ Authored an article on AI hallucinations in legal research, highlighting real-world risks of fabricated case law and the need for prompt precision. Advocated safe AI adoption in law through domain-specific models and ethical safeguards.